

10.4 The Medium is the Message (MDM)

Students learn about the physics involved in some modern communication and display techniques:

- fibre optics and exponential attenuation
- CCD imaging
- cathode ray tube
- liquid crystal and LED displays.

Exponential functions are used to model attenuation losses.

There are opportunities to develop information and communication technology skills using computer simulations.

In studying various types of display technology, students consider their relative power demands and discuss the choices made by organisations and by individuals.

Students will be assessed on their ability to:	Suggested experiments
83 draw and interpret diagrams using lines of force to describe radial and uniform electric fields qualitatively	
84 explain what is meant by an electric field and recognise and use the expression electric field strength $E = F/Q$	Demonstration of electric lines of force between electrodes
86 investigate and recall that applying a potential difference to two parallel plates produces a uniform electric field in the central region between them, and recognise and use the expression $E = V/d$	
87 investigate and use the expression $C = Q/V$	Use a Coulometer to measure charge stored

Students will be assessed on their ability to:	Suggested experiments
88 recognise and use the expression $W = \frac{1}{2} QV$ for the energy stored by a capacitor, derive the expression from the area under a graph of potential difference against charge stored, and derive and use related expressions, for example, $W = \frac{1}{2} CV^2$	Investigate energy stored by discharging through series/parallel combination of light bulbs
91 explore and use the terms magnetic flux density B , flux Φ and flux linkage $N\Phi$	
93 recognise and use the expression $F = Bqv \sin \theta$ and apply Fleming's left hand rule to charges	Deflect electron beams with a magnetic field
98 recall that electrons are released in the process of thermionic emission and explain how they can be accelerated by electric and magnetic fields.	